

Master's in Data Science and Advanced Analytics

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Deep Learning Project

Wikiart Image Classification Report

Group 3

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1 Introduction

This project focuses on developing a deep learning model to classify the author of a painting using a subset of the WikiArt dataset. This problem is particularly challenging due to high intra-class variability, strong visual similarity between artists, and class imbalance. Additionally, the problem is further complicated by the presence of artists with highly diverse styles (e.g., Picasso), making intra-class variability particularly challenging. These factors require careful model design and evaluation.

Throughout the project, multiple preprocessing strategies, model architectures, and training configurations were explored, including both convolutional neural networks trained from scratch and transfer learning approaches. A final ensemble method was also used to improve performance.

2 EDA and Preprocessing

The WikiArt dataset comprises 13 340 JPEG images (512×512 , RGB) spanning 23 artist classes. The EDA aims to uncover patterns and insights to inform model development.

To ensure data integrity, near-duplicate images were identified using the difference hash [4] algorithm, which encodes brightness gradients between adjacent pixels into a binary fingerprint. Pairs with a [9] distance ≤ 5 were considered near-duplicates, leading to the exclusion of **66 images** to prevent data leakage. No corrupted, mislabeled, or anomalous images were found.

The data set was then split into train, validation and test sets following a **70–15–15 ratio**, stratified per artist class. This split was performed once, so that images were physically separated into dedicated folders, guaranteeing consistency during multiple models trainings.

Despite using stratified splitting, class imbalance persisted among the artists. Manual class weights were tested to counter this, but they were applied selectively only to the models where they demonstrably improved results.

After verifying that the class distribution matched across the three partitions, a **data augmentation pipeline** was defined, applied consistently to all images in the train set. Said pipeline comprises random horizontal flips, rotations ($\pm 10\%$), zoom ($\pm 15\%$), and contrast perturbation ($\pm 15\%$).



Figure 1: Example of training images, illustrating the effect of our chosen augmentations

Interestingly, we observed that data augmentation was highly dependent on input resolution. At lower resolutions (128×128), augmentation degraded performance by removing fine-grained details such as brushstrokes. However, at higher resolutions (224×224), it improved generalization, suggesting that preserving spatial detail is critical for this task.

3 Methodology

In this section, we will discuss the process of designing and training different Convolutional Neural Networks to predict the author of the painting.

3.1 Scratch Models

3.1.1 Baseline Model

To establish a reference point, we first implemented a deliberately simple CNN baseline. Its purpose was methodological: any deeper architecture or transfer learning approach should clearly outperform it to justify the additional computational cost. The model was intentionally minimal, consisting of two convolutional blocks with 32 and 64 filters, followed by Global Average Pooling and a Dense layer with 64 units before the 23-class softmax output. Given the difficulty of the WikiArt task, the baseline was not expected to achieve strong results, but rather to provide a lower bound for accuracy and macro F1. In this sense, it served as a useful diagnostic benchmark for all subsequent experiments.

3.1.2 Optuna Search

We choose to use Optuna [1], a automatic hyperparameter optimization framework to find hyperparameters. Each trial consisted of training a candidate CNN architecture with early stopping (monitoring validation F1), a minimum improvement threshold of 0.005, and the learning rate reduction on plateau callback was also applied.

Several iterations were needed to refine the search space while keeping the optimization computationally feasible. To ensure architectural validity, monotonicity constraints were manually imposed on both convolutional filter counts and dense layer units.

The full search space is summarised in Table 3. All trials were logged to a persistent CSV file and a SQLite database, allowing the study to be interrupted and resumed without loss of progress. However, when any search space parameter was modified between runs, both the database and CSV were constantly deleted to avoid corrupting the TPE model with incomparable trial scores.

The search space and number of epochs were designed to balance model expressiveness with computational cost. Convolutional dropout was included as a regularization mechanism, while weight decay was tuned jointly with the learning rate to balance parameter shrinkage and optimization dynamics.

A table comparison of the trials can be found in the Table 4 in the appendix. From there, to select the most promising trial for fine-tuning, we decided to identify a trial that did not stop early (i.e., completed all 35 epochs) and achieved the highest validation accuracy

and F1 score. The early stopping criterion is important because models that stop before reaching the full 35 epochs had likely already reached a performance plateau.

We then aimed at improving the best trial, 21, retraining that model from the beginning, and fine-tuning with a smaller learning rate.

3.1.3 Residual Blocks Model

A closer analysis of the trial results revealed that the architectures explored by Optuna were unable to capture the underlying complexity of the data. To address this limitation, we decided to investigate deeper and more expressive model architectures.

However, due to computational constraints, it was not feasible to extend the Optuna search space to include larger configurations, particularly with respect to the number of layers.

We conducted a series of manual experiments to progressively improve the model, with some of the most relevant steps summarised in Table 1.

Table 1: Summary of residual model progression

Version	Key Change	Val F1	Test F1
Baseline	Simple CNN (2 blocks, 224×224)	~0.25	0.24
v3	+ Label smoothing	~0.44	0.45
v5a	4 blocks + regularization	~0.61	0.61
v5c	Higher resolution (224×224)	~0.66	0.64
v5e	+ Augmentation + tuning	~0.67	0.67
v5d	+ number of epochs	~ 0.72	0.70

Early results indicated that increasing model capacity alone was not sufficient, as deeper configurations without proper regularization led to overfitting. The introduction of label smoothing (v3) provided a clear improvement, highlighting the importance of regularization. The most significant performance gain was achieved with deeper architectures combined with regularization (v5a), showing that balancing capacity and overfitting is critical.

Further improvements were achieved by increasing the input resolution to 224×224, confirming the importance of fine-grained visual features for this task. In contrast, some choices, such as early data augmentation or excessive model complexity, led to unstable or degraded results. Later refinements produced only modest gains, suggesting diminishing returns and highlighting the limitations of training from scratch, which motivated the transition to transfer learning.

3.2 Transfer Learning Models

3.2.1 VGGNet, EfficientNetB3 and EfficientNetB5

In our initial transfer learning experiments, we evaluated three ImageNet pre-trained architectures: VGG16 [5], EfficientNetB3, and EfficientNetB5 [6]. All were adapted to

the 23-class artist classification task with a custom prediction head and trained using the same two-phase strategy: first with the backbone frozen, and then with partial fine-tuning of the final layers. To improve stability and reduce overfitting, we introduced smaller batch sizes, class weights, and a callback configuration based on validation macro F1. This shared setup allowed a fair comparison of the backbone architectures while keeping the rest of the pipeline largely consistent.

3.2.2 ResNet50

We adopted a transfer learning approach based on the ResNet50 [7] architecture pre-trained on ImageNet. The use of ImageNet weights was motivated by insights from the EDA. Since these features are well captured by models trained on large-scale natural image datasets, leveraging pre-trained weights provided a strong starting point compared to training from scratch. The model was adapted to the 23-class classification task by replacing the original classification head with a custom architecture. This head combines Global Average Pooling and Global Max Pooling, followed by batch normalization, fully connected layers, and dropout regularization. This design allows the model to capture both global stylistic patterns and localized discriminative features, which are particularly relevant for artistic image classification. Training was conducted in two phases. In the first phase, the backbone was frozen and only the classification head was trained. In the second phase, partial fine-tuning was applied by unfreezing the deeper layers of the network while keeping Batch Normalization layers frozen to preserve stable feature statistics and prevent training instability. To further improve generalization, several regularization and optimization techniques were incorporated, including label smoothing, MixUp data augmentation, and the AdamW optimizer. The training process was monitored using the validation macro F1-score, which was also used as the primary criterion for early stopping and learning rate scheduling.

3.2.3 EfficientNetV2S

In addition to ResNet50, we explored a transfer learning approach using EfficientNetV2-S [8] pre-trained on ImageNet. A similar training pipeline and architectural design principles were adopted, including the use of pre-trained weights, a custom classification head, and a two-phase training procedure with a frozen backbone followed by partial fine-tuning. The classification head followed the same design, combining Global Average Pooling and Global Max Pooling, followed by batch normalization, dense layers, and dropout regularization. Batch Normalization layers were also kept frozen during fine-tuning to ensure training stability. Despite using a comparable training setup, EfficientNetV2-S consistently outperformed ResNet50, achieving an improvement of approximately 3–4 percentage points in macro F1-score. This improvement can be attributed to the more advanced architecture of EfficientNet, which is designed to achieve better parameter efficiency and stronger feature extraction through compound scaling of depth, width, and resolution.

3.3 Key Design Choices

The most impactful design choices identified throughout the project are summarised below:

- **Transfer Learning with ImageNet:** ImageNet pre-trained weights provided a strong initialization and clearly outperformed models trained from scratch, showing the value of transfer learning for artistic image classification.
- **Input Resolution:** Increasing the input resolution from 128×128 to 224×224 consistently improved performance by preserving fine-grained details such as brush-strokes and texture patterns.
- **Dual Pooling Classification Head:** Combining Global Average Pooling (GAP) and Global Max Pooling (GMP) helped capture both global stylistic patterns and highly discriminative local features.
- **Frozen Batch Normalization Layers:** Keeping Batch Normalization layers frozen during fine-tuning improved training stability and preserved the pre-trained feature statistics.
- **Regularization Techniques:** Label smoothing, MixUp, and dropout improved generalization and helped reduce overfitting.
- **Test-Time Augmentation (TTA):** Applying TTA at inference time by averaging predictions over multiple augmented versions of each image resulted in a significant performance gain [2]. This indicates that prediction aggregation improves robustness to small input variations. However, there are no free lunches, and this technique also destroys the performance of some models. Further details on this approach are provided in Appendix X (see Table 5).

These design choices were validated through empirical experimentation and collectively contributed to the final performance improvements observed.

4 Evaluation

The main evaluation metric in this project is the **macro F1-score**, since the dataset is imbalanced and some artists are much more represented than others. Unlike accuracy, macro F1 gives equal importance to all classes, making it a fairer metric for model selection in this multi-class setting. Accuracy is also reported as a complementary measure. The final ensemble, combining ResNet50 and EfficientNetV2-S, achieved the best result of the project, improving macro F1 by about 3 percentage points and reaching 0.87.

5 Results and Analysis

5.1 Model Comparison

Table 2: Comparison of all evaluated models using the main performance metrics.

Model	Val. Accuracy	Val. Macro F1	Test Accuracy	Test Macro F1
Baseline Model	0.3409	0.2599	0.3042	0.2341
Optuna (Trial 21)	0.5399	0.5271	0.5258	0.5142
Residual Block	0.7439	0.7188	0.7266	0.7019
VGGNet	0.7010	0.6750	0.7102	0.6941
EfficientNetB3	0.7263	0.7067	0.7247	0.7089
EfficientNetB5	0.7505	0.7363	0.7490	0.7381
EfficientNetV2S	0.8389	0.8167	0.8484	0.8294
ResNet50	0.8051	0.7889	0.8067	0.7918

5.2 Grad-CAM Visualizations

The Grad-CAM[3] visualizations 6 support the EDA findings by showing that the models focus on meaningful artistic features, such as texture, composition, and illumination, rather than irrelevant background regions. In works by Pablo Picasso, Albrecht Dürer, and Rembrandt, the strongest activations appear on stylistically distinctive elements, suggesting that the models capture relevant visual patterns associated with each artist. The visualizations also highlight one of the main challenges of the dataset: distinguishing between artists from closely related movements. For example, in a Claude Monet painting, EfficientNetV2S focuses on the characteristic atmospheric brushstrokes, while ResNet50 misclassifies the work as Childe Hassam, indicating that the error stems from genuine stylistic overlap. Overall, these results suggest that the models learned meaningful representations and support the use of an ensemble and test-time augmentation to handle subtle ambiguities.

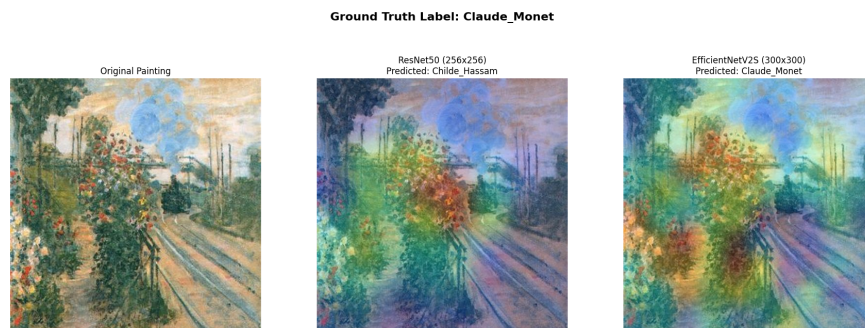


Figure 2: Claude Monet - Effnet vs ResNet

6 Discussion, Key Findings and Future Work

A central finding concerns the role of data augmentation. At low resolution (128×128), augmentation degraded performance due to the loss of fine-grained details such as brush-strokes. In contrast, at higher resolution (224×224), augmentation improved generalization and reduced the train-validation gap, highlighting the importance of preserving spatial detail in artistic image classification.

Regularization strategies also showed important interactions. While label smoothing and class weighting individually improved performance, their combination led to worse results, suggesting overlapping effects that weaken the learning signal. Additionally, SpatialDropout2D reduced overfitting by closing the train-validation gap, although it did not significantly increase overall performance, indicating that the model had reached its capacity when trained from scratch.

From an architectural perspective, increasing depth without proper regularization led to overfitting, while alternative design choices such as replacing MaxPooling with strided convolutions had negligible impact. Input resolution proved to be a critical factor, with a consistent improvement of approximately 3 p.p in macro F1 when increasing from 128×128 to 224×224 .

The transition to transfer learning resulted in the most significant performance gains. Freezing Batch Normalization layers during fine-tuning ensured training stability, while monitoring validation macro F1 instead of validation loss led to better checkpoint selection. Among all techniques, Test-Time Augmentation (TTA) produced the largest improvement, increasing macro F1 by approximately 4.9 percentage points. This result suggests a distribution mismatch between training and inference and highlights the effectiveness of prediction aggregation.

Future work could explore more advanced ensemble methods, hierarchical or multi-stage classification for the most difficult classes, the incorporation of metadata, newer architectures such as Vision Transformers, and deeper interpretability analyses to better understand model behavior and guide further improvements.

7 Conclusion

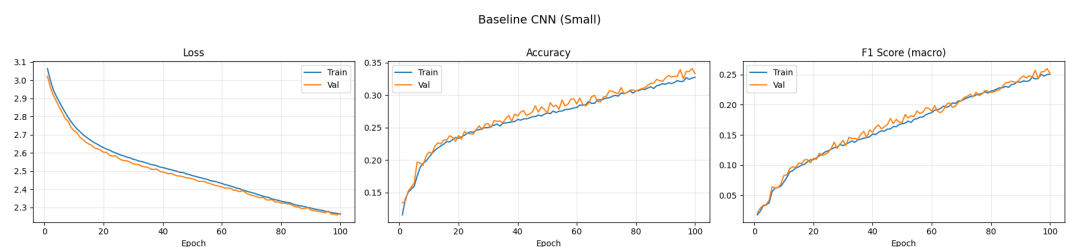
This project showed that artist classification on the WikiArt dataset is a challenging task due to class imbalance, strong stylistic overlap, and high intra-class variability. Our experiments demonstrated that transfer learning clearly outperformed models trained from scratch, with the best results obtained using EfficientNetV2-S and ResNet50-based pipelines. We also found that input resolution, careful regularization, frozen Batch Normalization during fine-tuning, and evaluation strategies such as Test-Time Augmentation played an important role in improving generalization. Overall, the results confirm that combining strong pre-trained backbones with well-designed training and evaluation procedures is an effective approach for fine-grained artistic image classification.

References

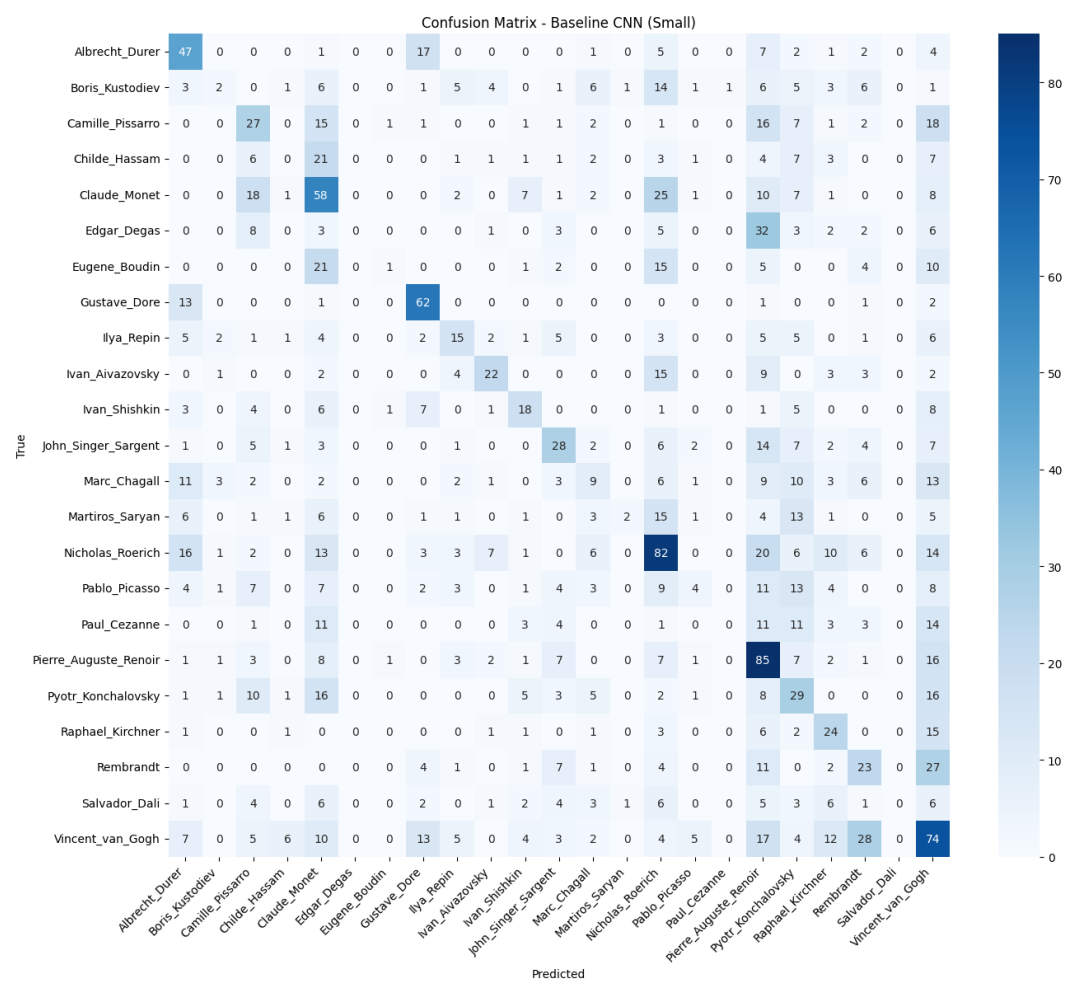
- [1] Akiba, T. et al. (2019). *Optuna: A Next-generation Hyperparameter Optimization Framework*. <https://optuna.org>
- [2] Shanmugam, D. et al. (2020). *Better Aggregation in Test-Time Augmentation*. arxiv.org/abs/2011.11156
- [3] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*. Proceedings of the IEEE International Conference on Computer Vision (ICCV). arxiv.org/abs/1610.02391
- [4] Buchner, J. (2025). ImageHash (Version 4.3.2) [Computer software]. PyPI. <https://pypi.org/project/ImageHash/>
- [5] Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. Preprint arXiv:1409.1556.
- [6] Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. In *Proceedings of the 36th International Conference on Machine Learning* (ICML 2019, PMLR 97, pp. 6105–6114).
- [7] He, K., Zhang, X., Ren, S., & Sun, J. (2015). *Deep Residual Learning for Image Recognition*. .
- [8] Tan, M., & Le, Q. V. (2021). *EfficientNetV2: Smaller Models and Faster Training*. In *Proceedings of the 38th International Conference on Machine Learning* (ICML 2021).
- [9] Hamming, R. W. (1950). *Error Detecting and Error Correcting Codes*. Bell System Technical Journal.

Appendix A: Scratch Models — Confusion Matrices and Training Histories

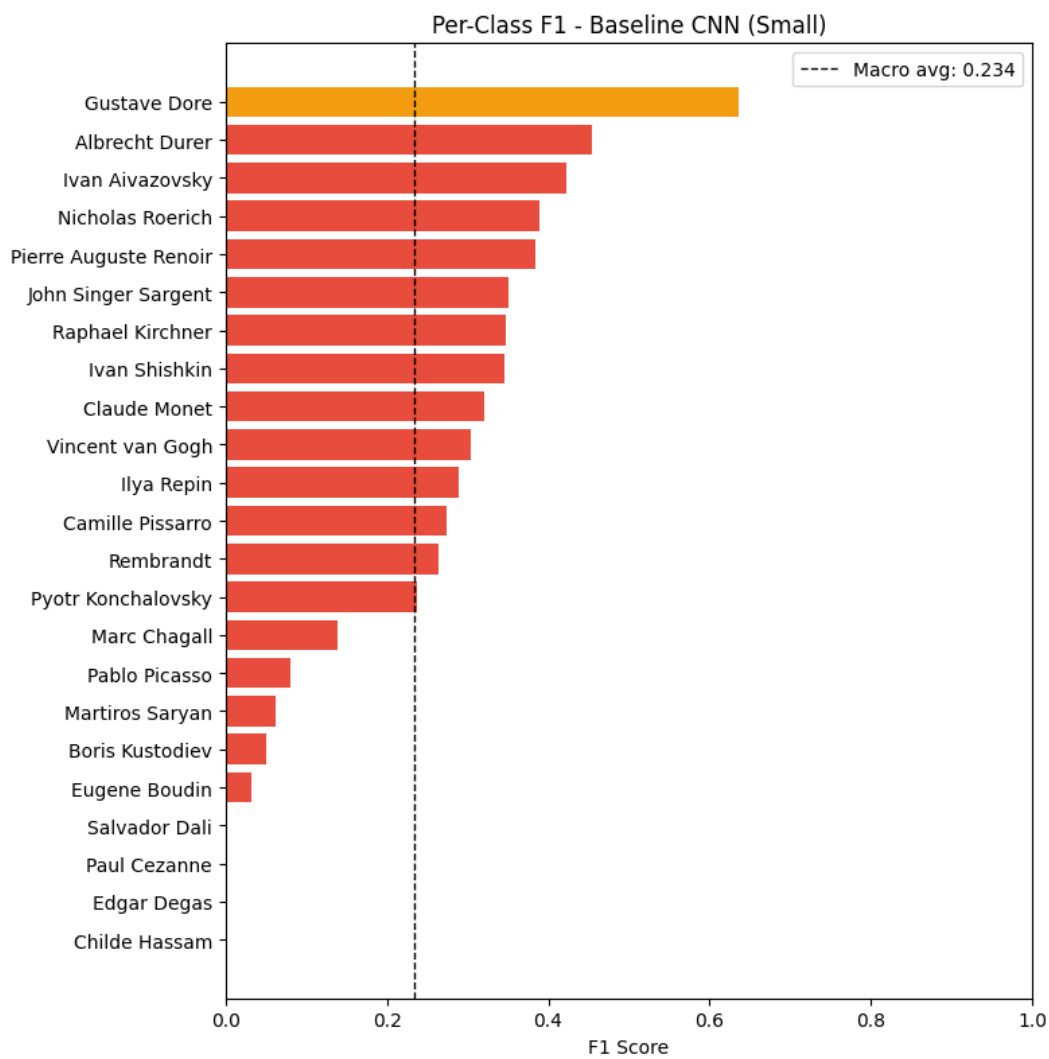
A.1 Baseline Model



Model Evolution

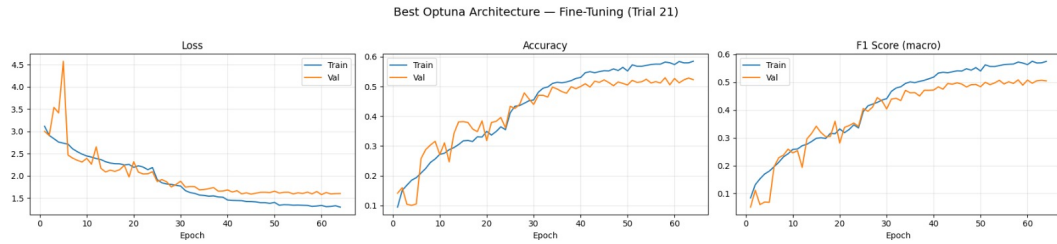


Confusion Matrix

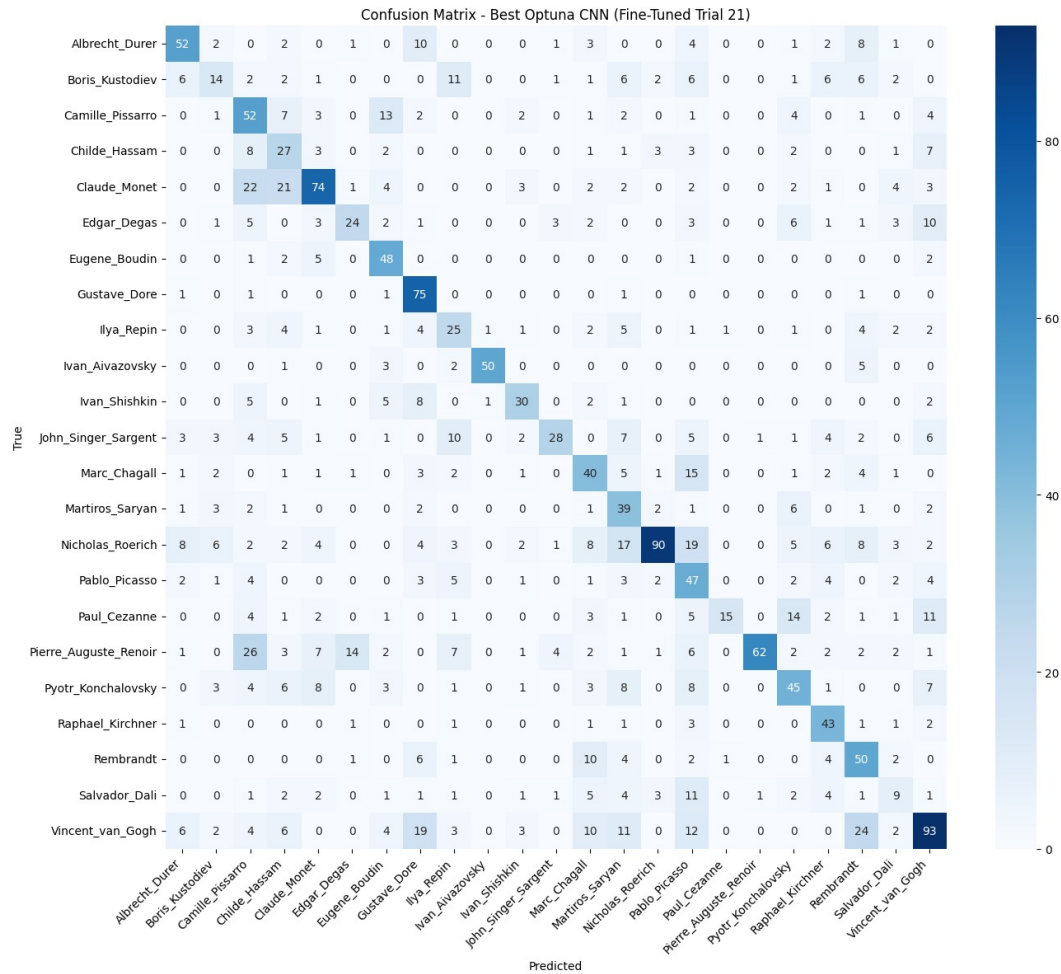


Training History

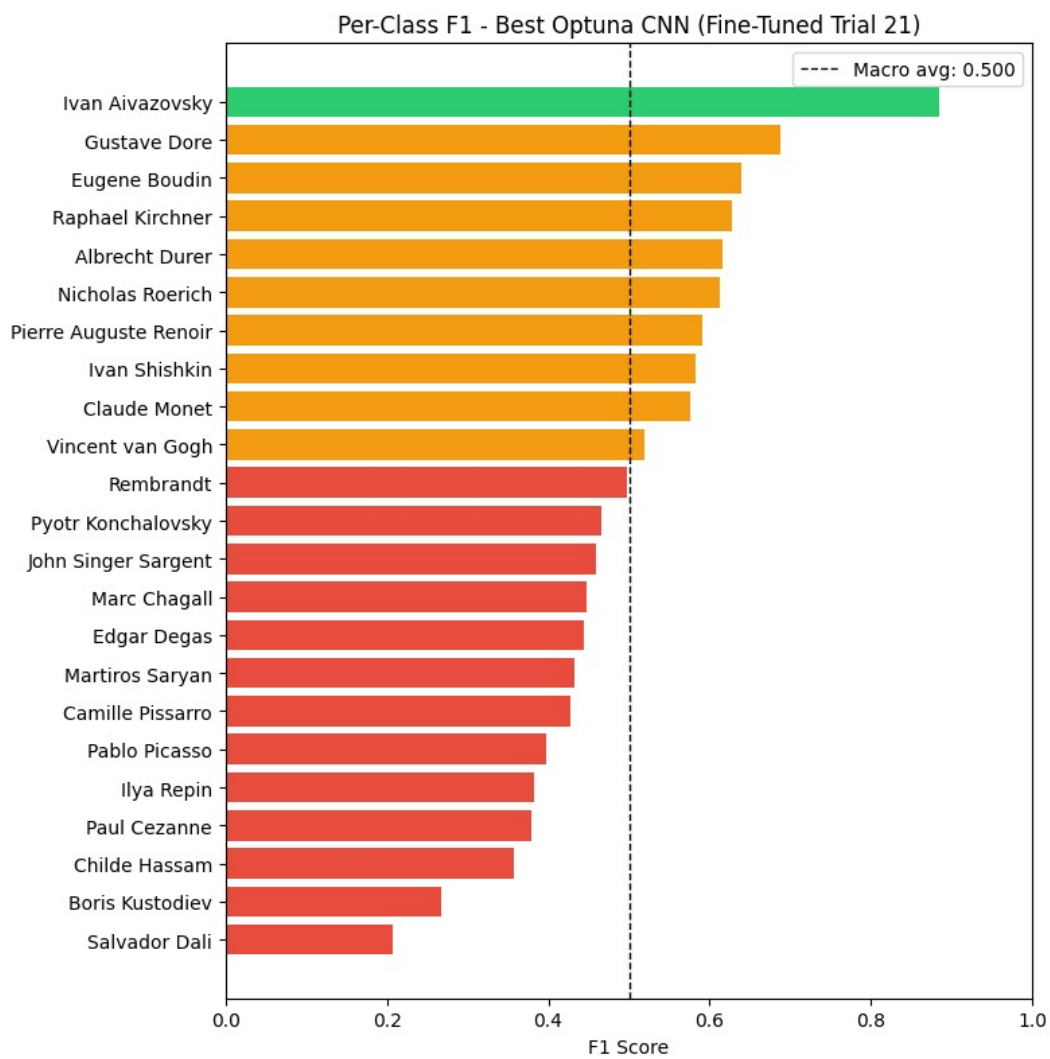
A.2 Optuna Trial 21



Model Evolution

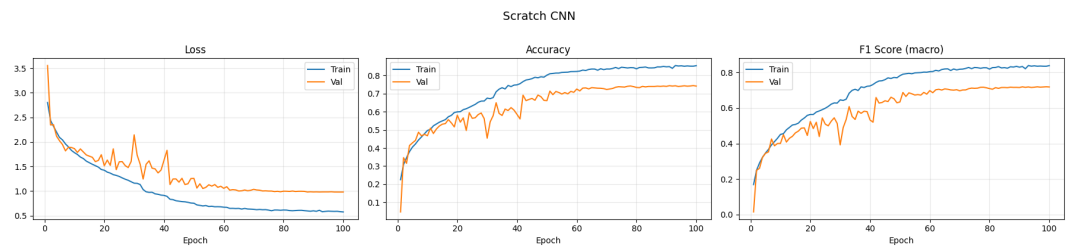


Confusion Matrix

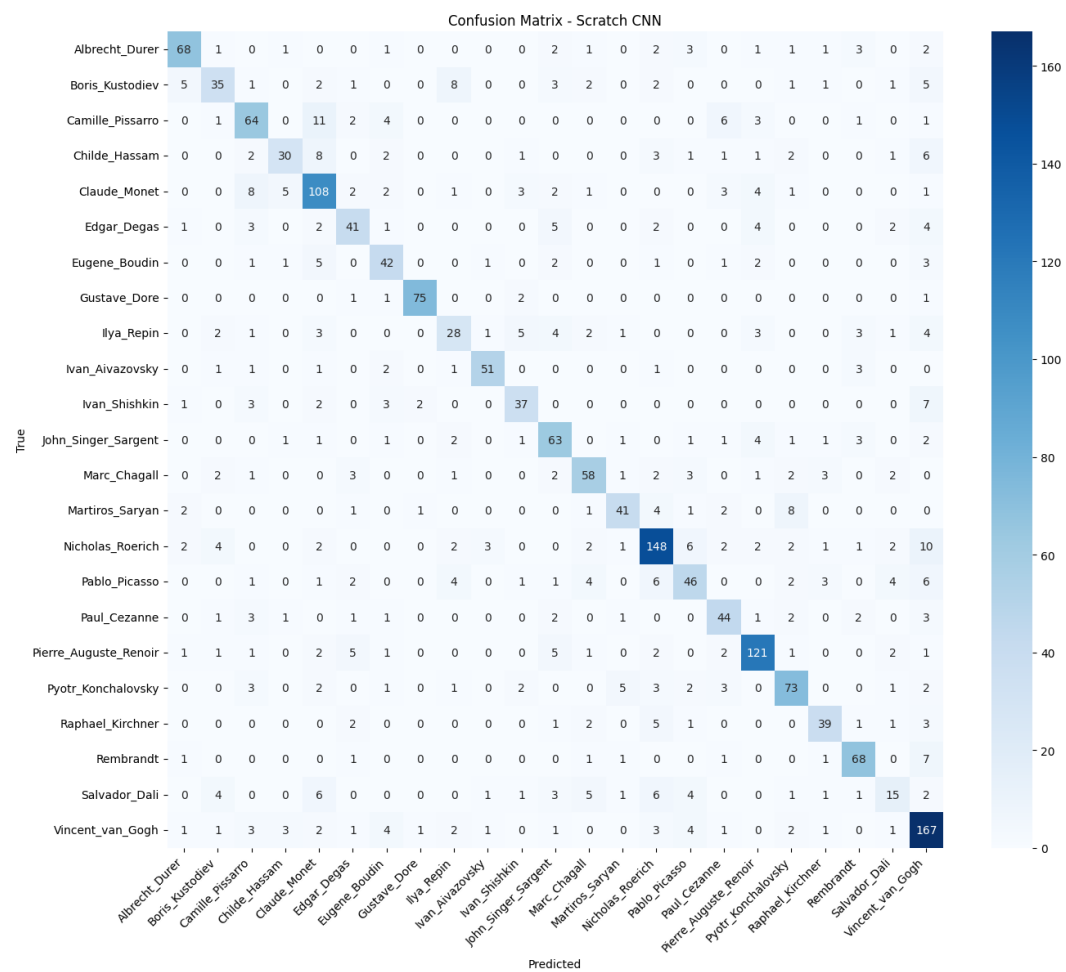


Training History

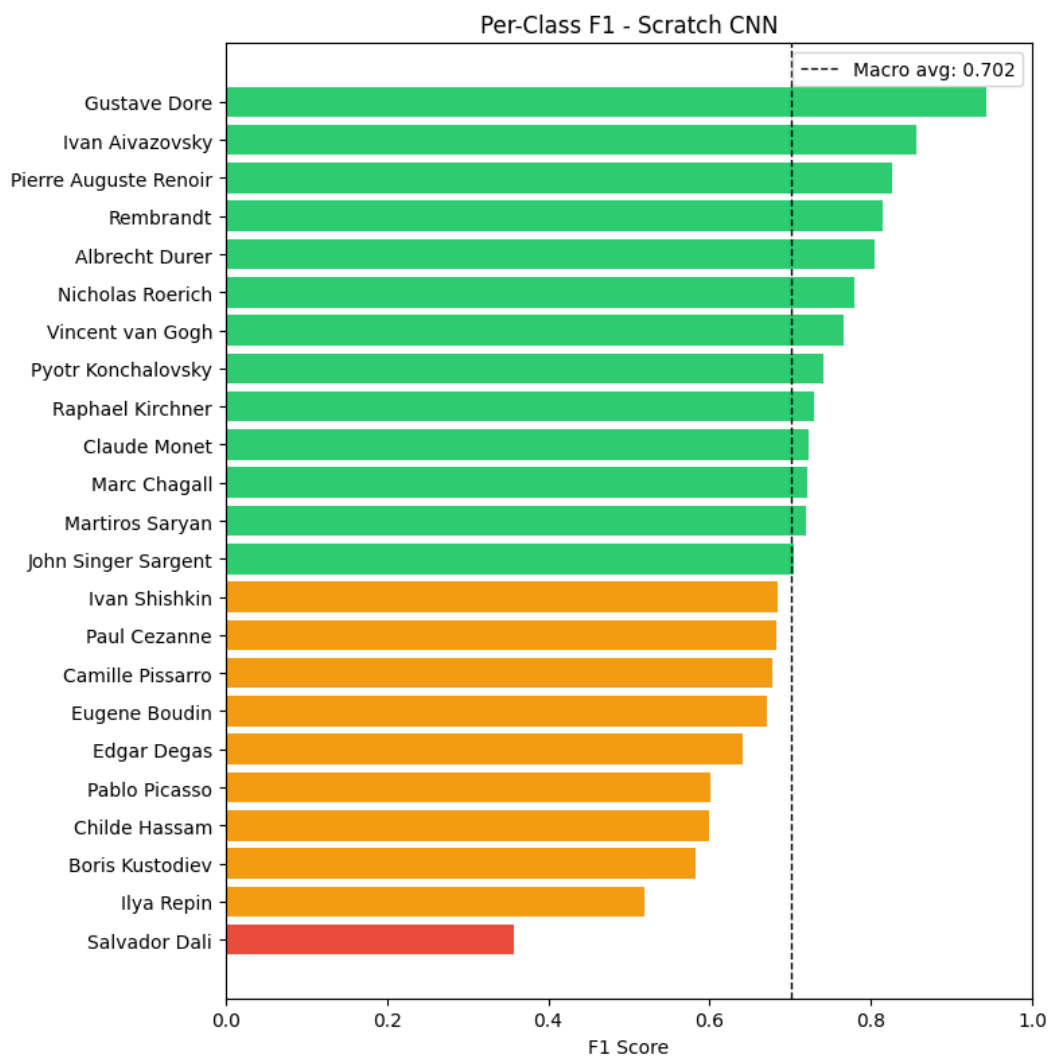
A.3 Residual Block Model



Model Evolution



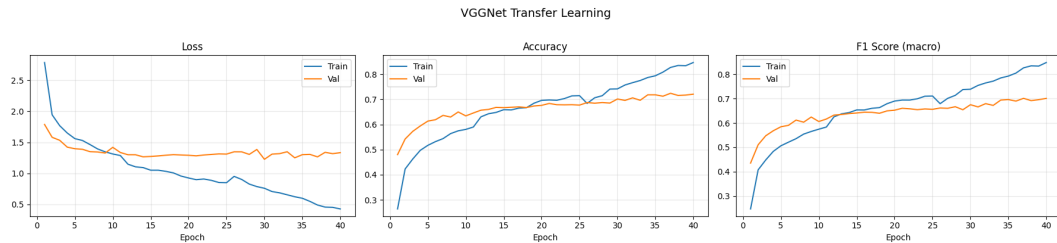
Confusion Matrix



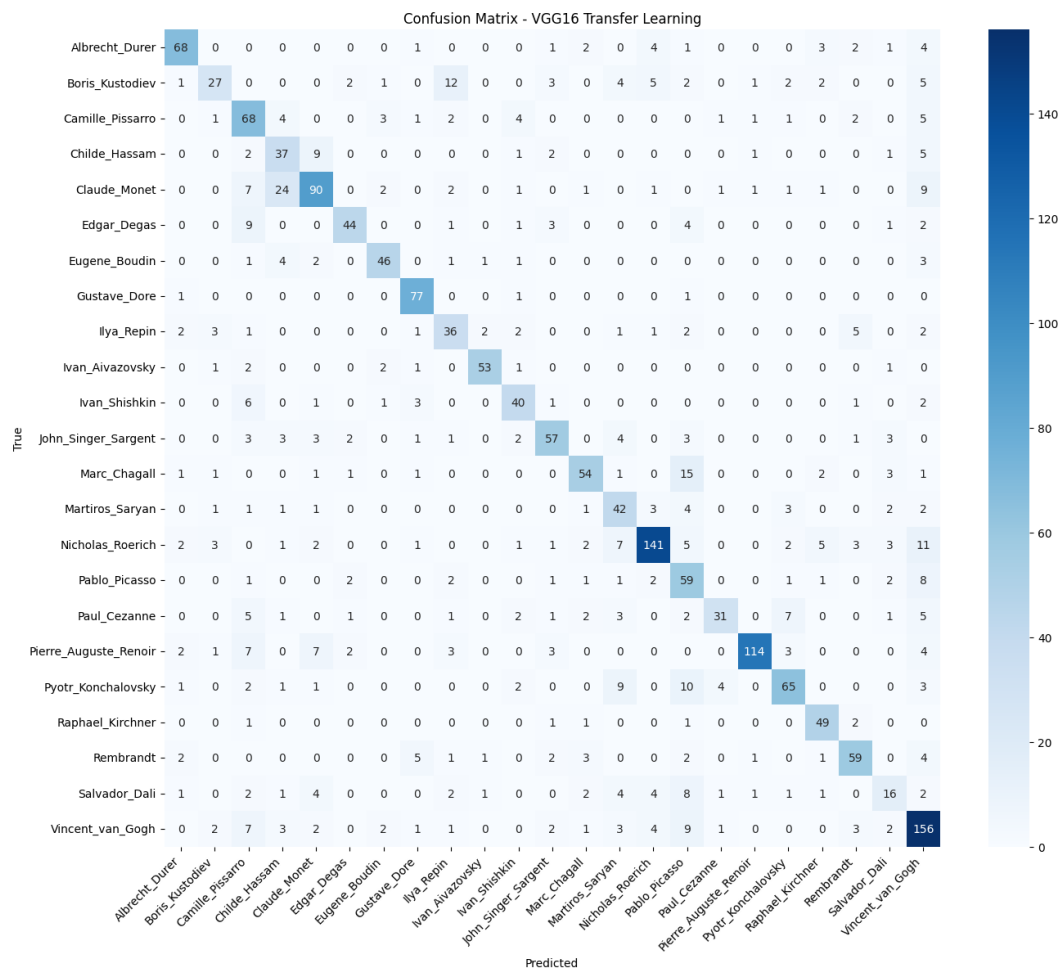
Training History

Appendix B: Transfer Learning Models — Confusion Matrices and Training Histories

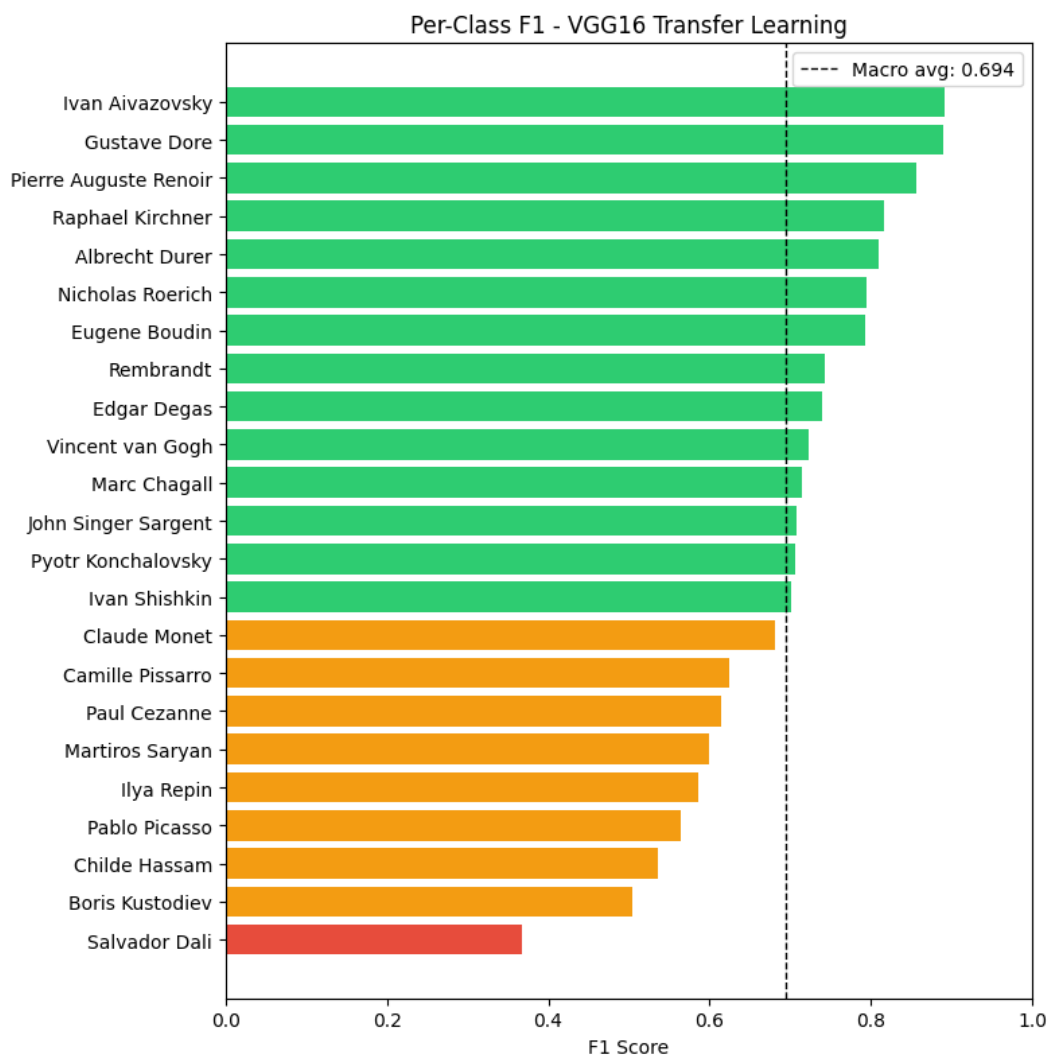
B.1 VGGNet



Model Evolution

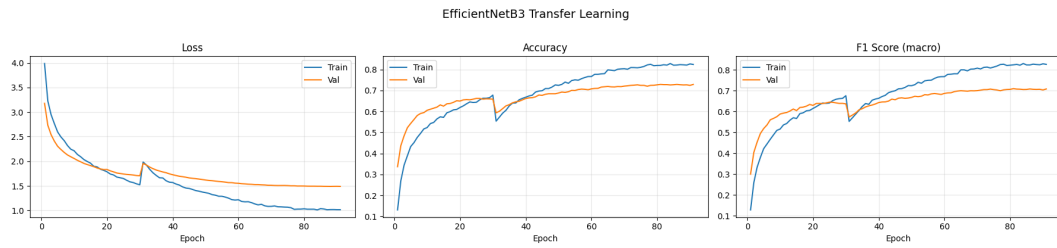


Confusion Matrix

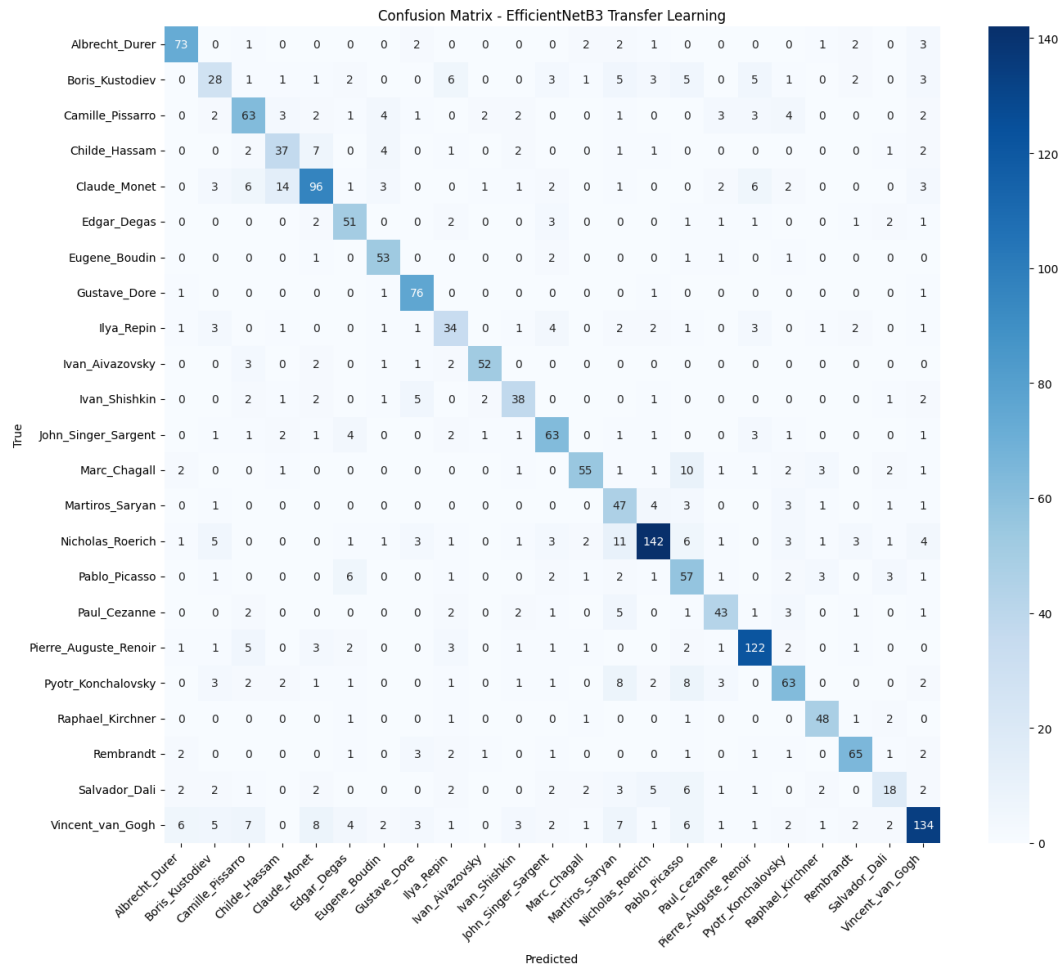


Training History

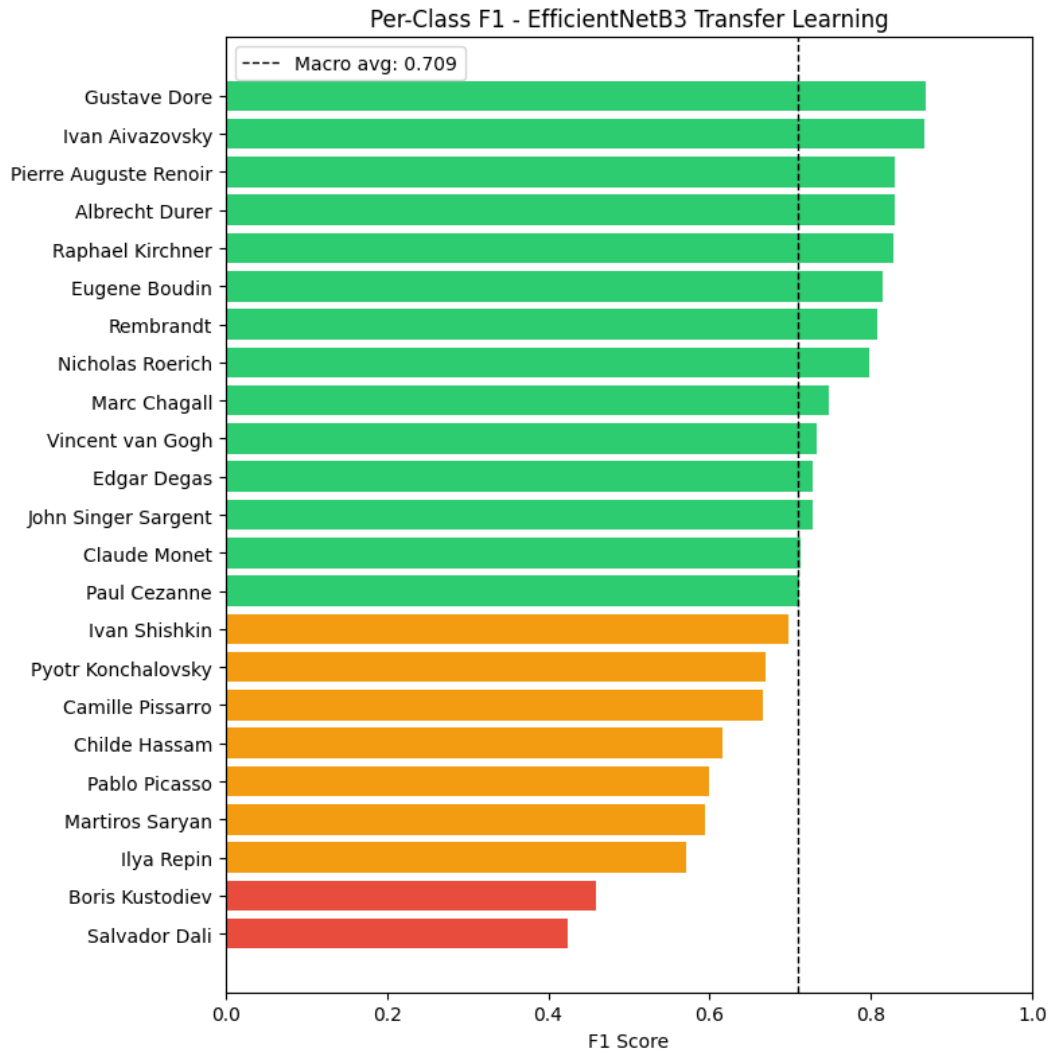
B.2 EfficientNetB3



Model Evolution

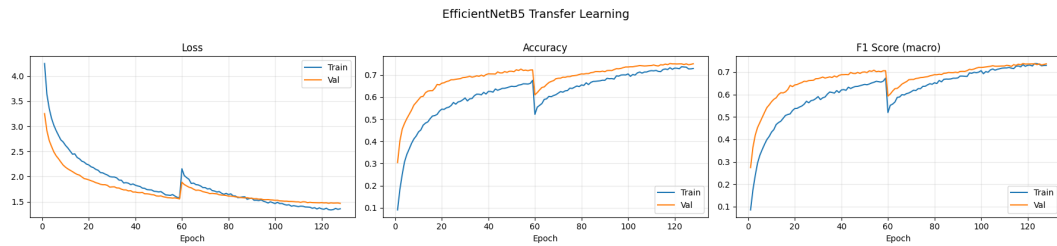


Confusion Matrix

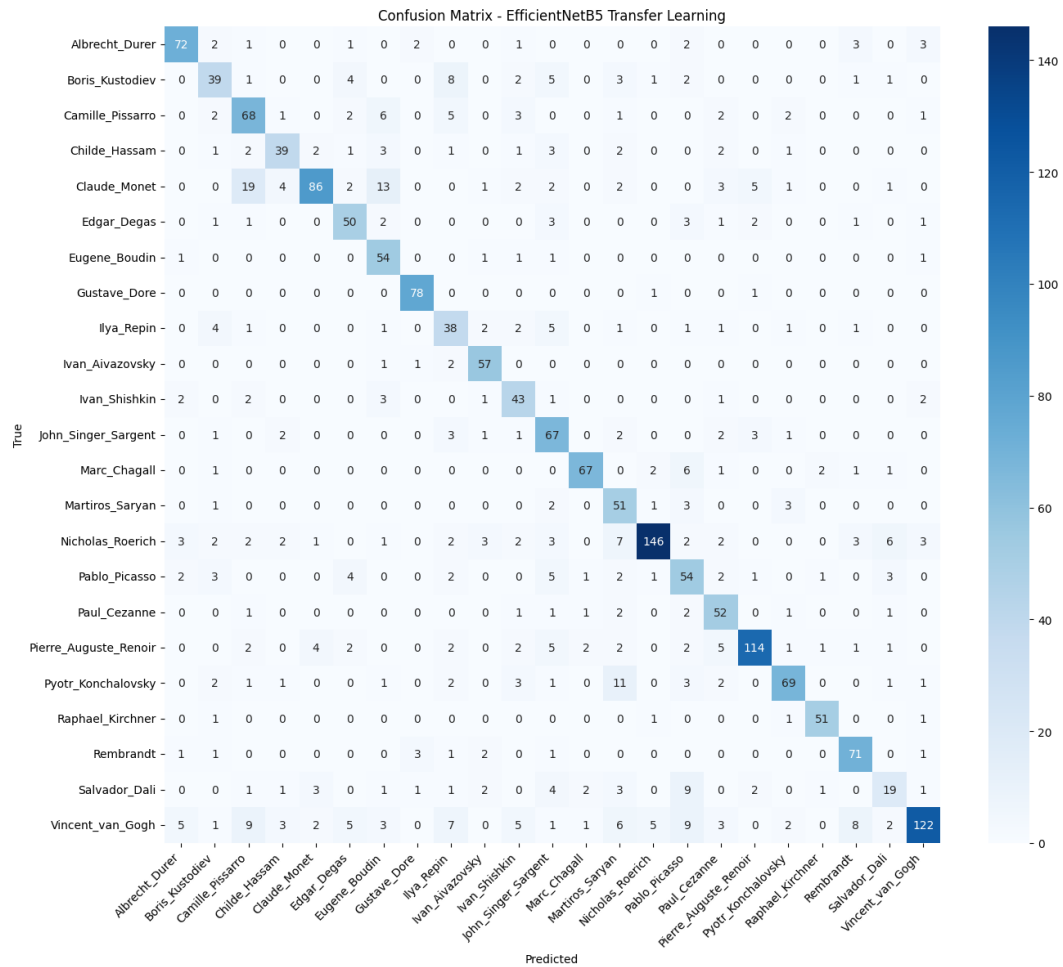


Training History

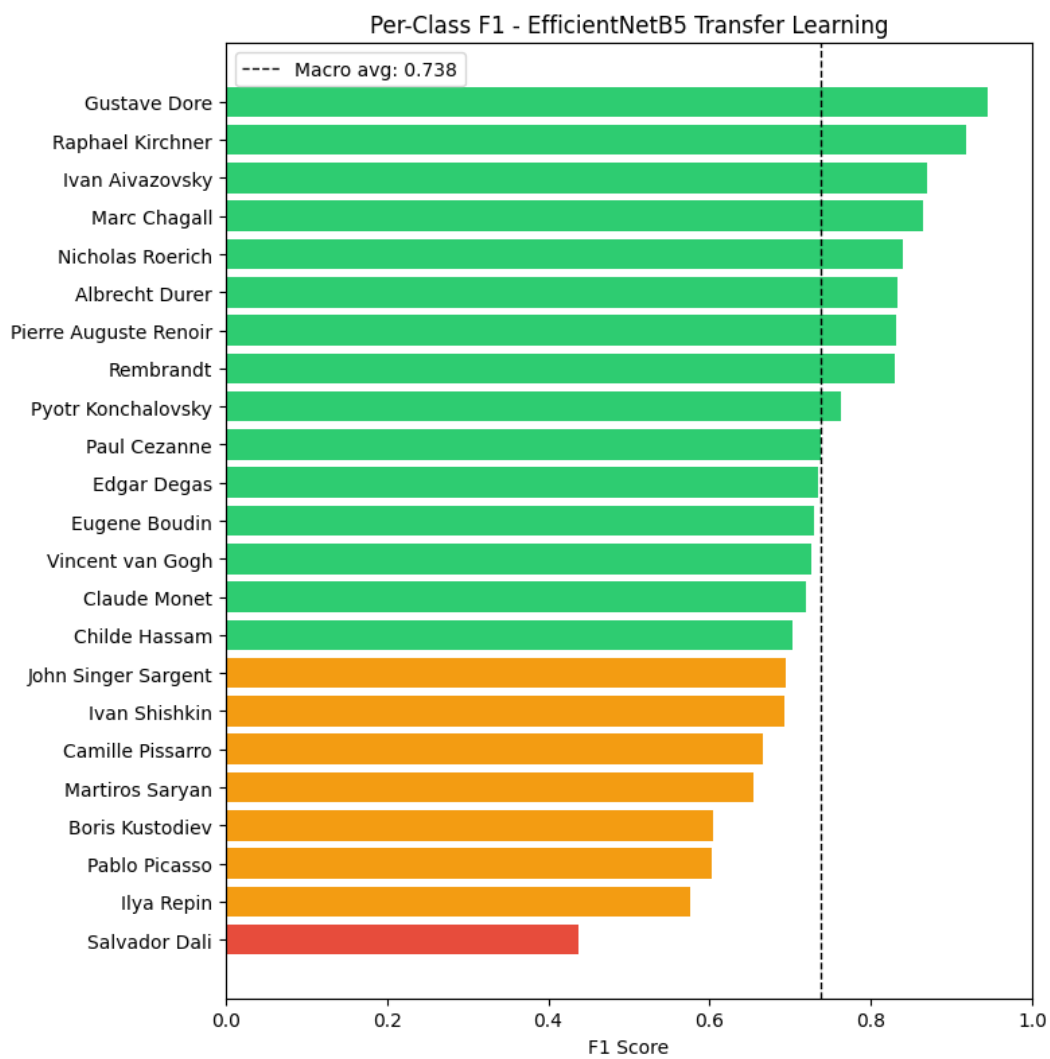
B.3 EfficientNetB5



Model Evolution

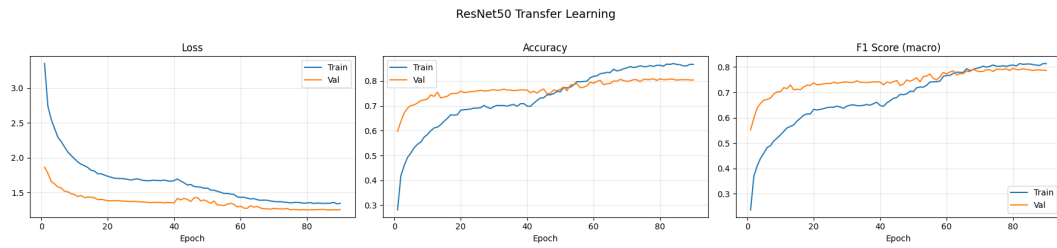


Confusion Matrix

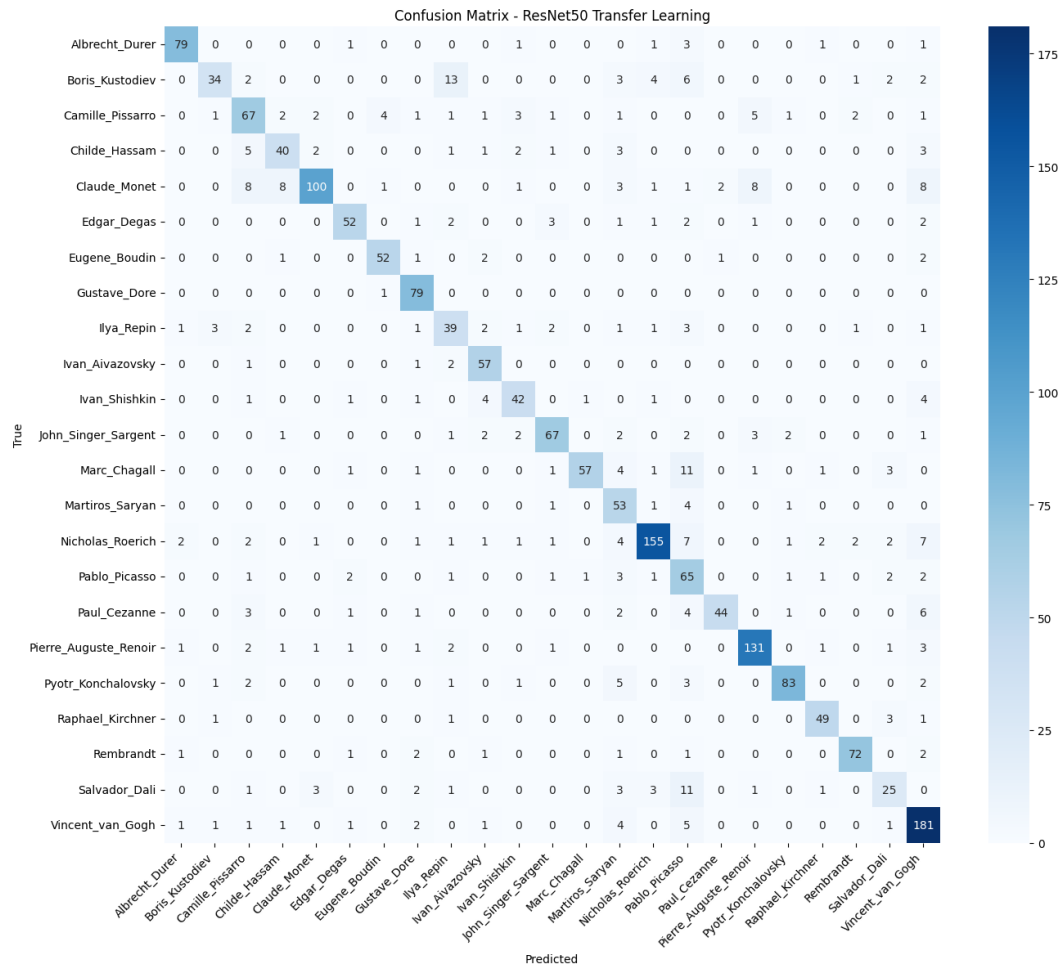


Training History

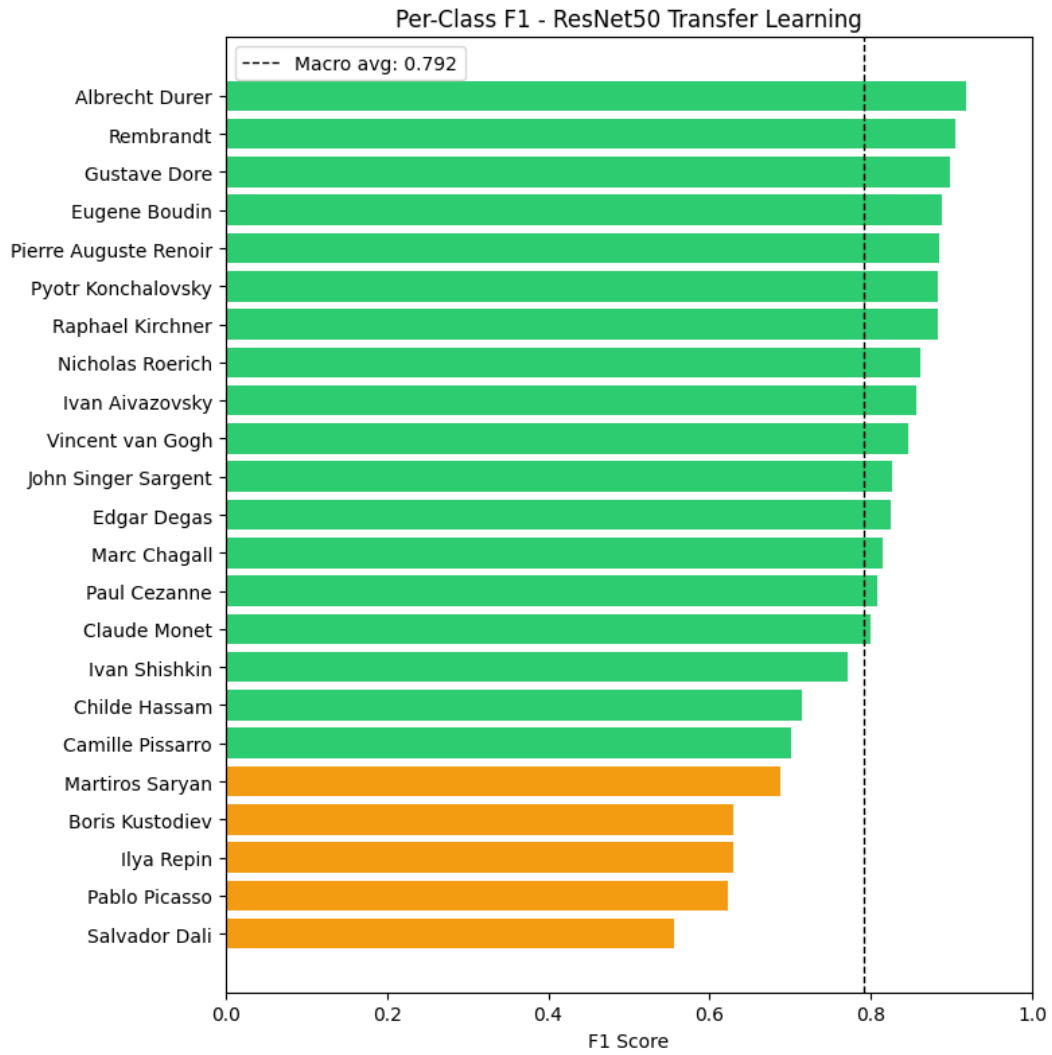
B.4 ResNet50



Model Evolution

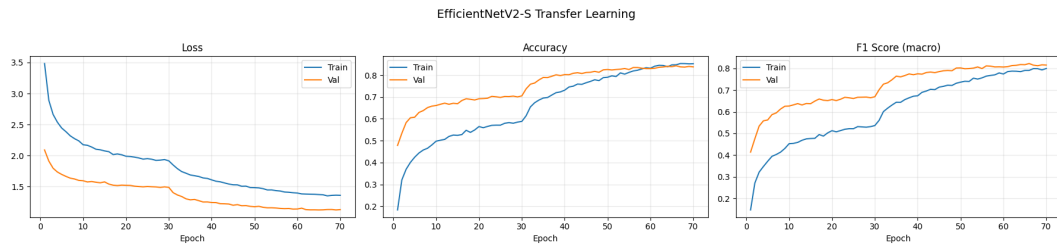


Confusion Matrix

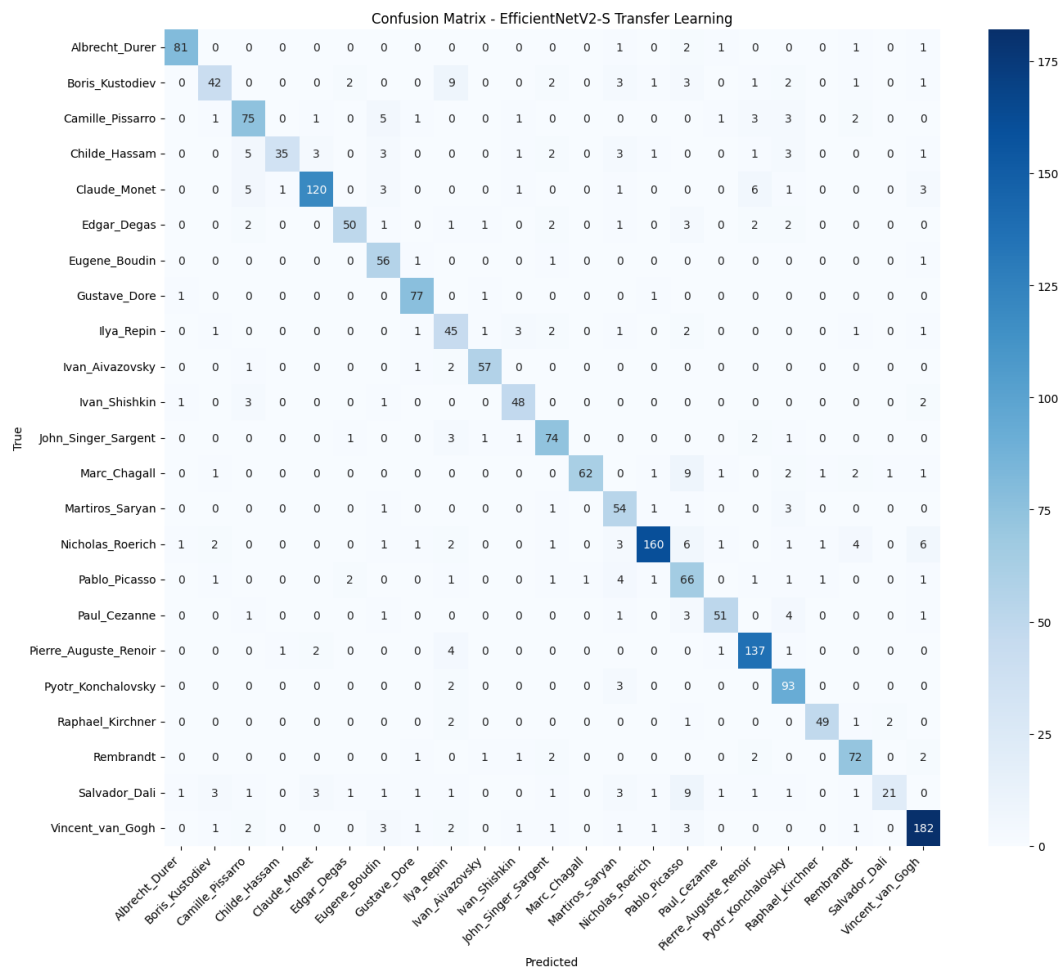


Training History

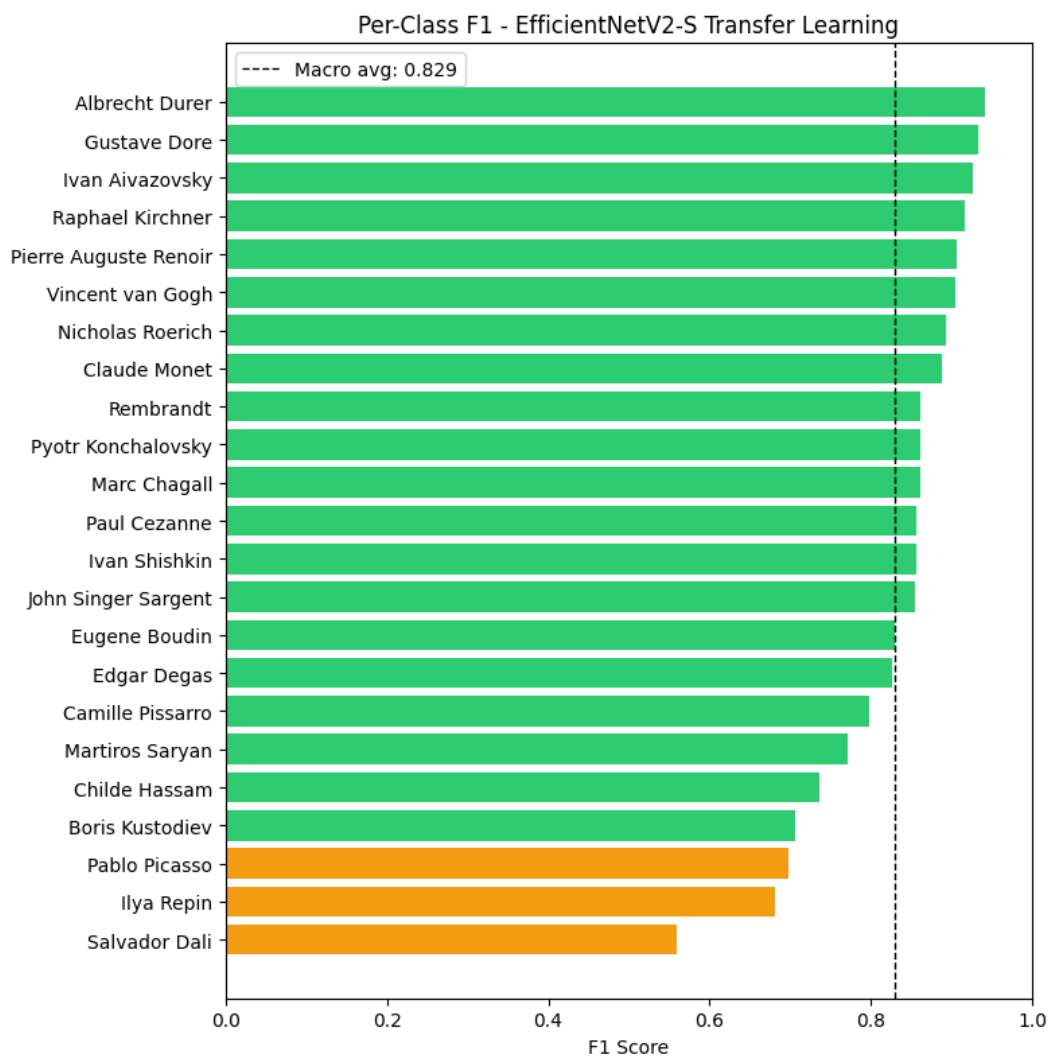
B.5 EfficientNetV2S



Model Evolution

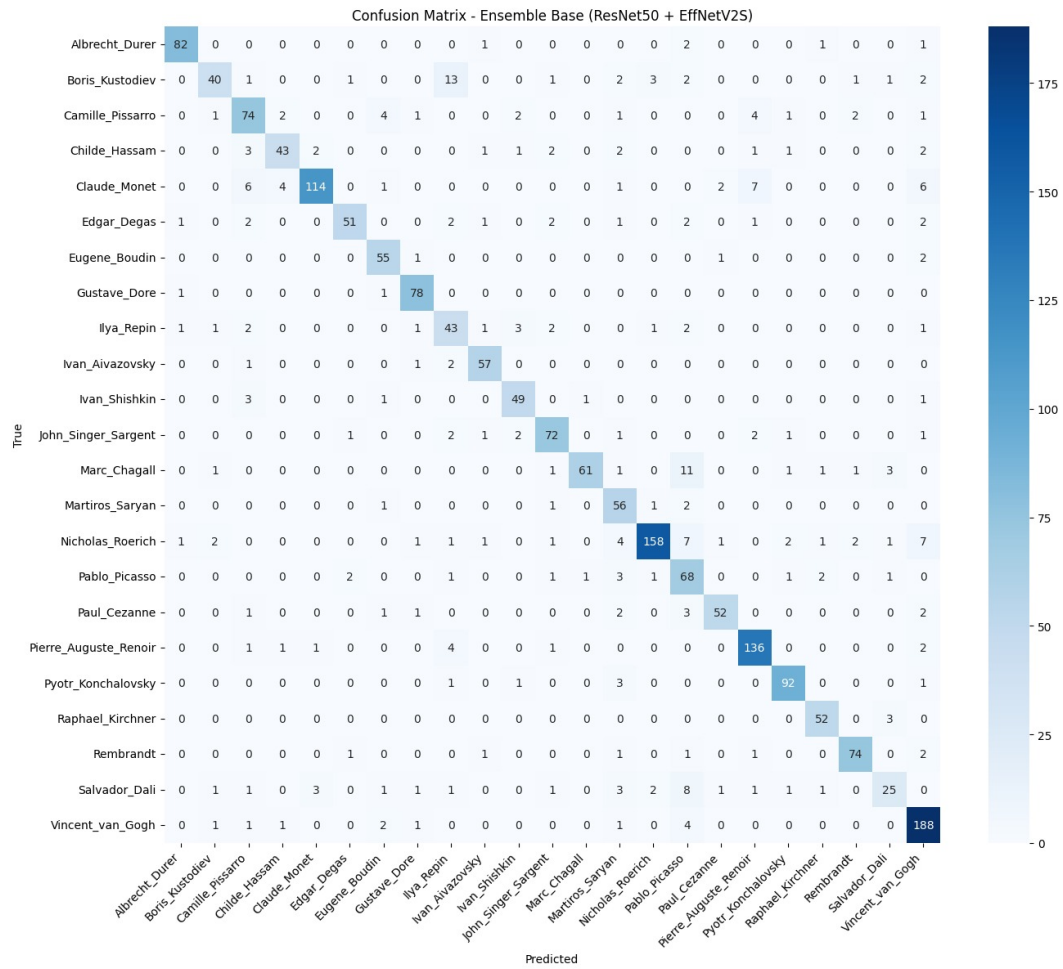


Confusion Matrix

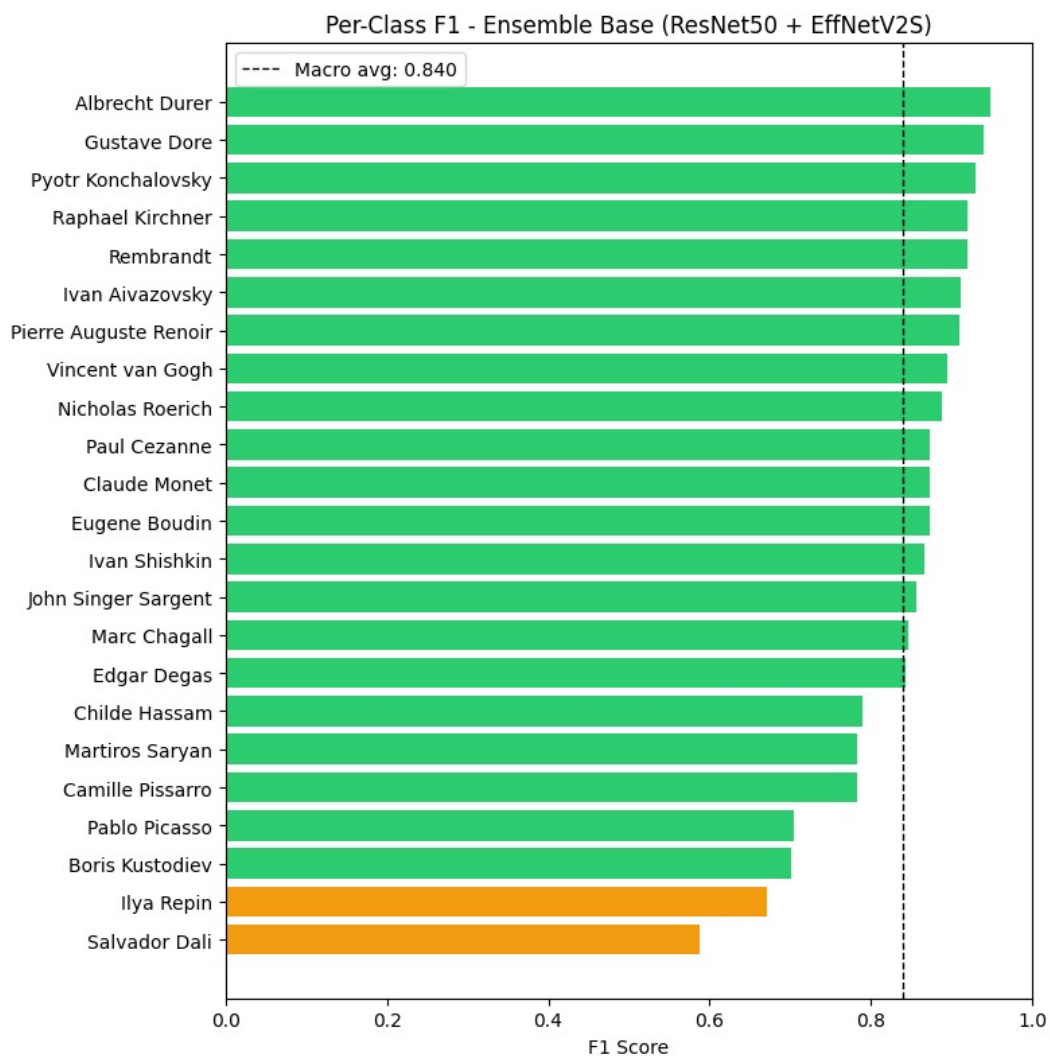


Training History

B.6 Ensemble Results



Model Evolution



Training History

Appendix C: Optuna Trials

Table 3: Hyperparameter Search Space

Component	Hyperparameter	Range / Values
Training	Learning rate	$[5 \times 10^{-4}, 5 \times 10^{-3}]$ (log scale)
	Weight decay	$[10^{-4}, 10^{-2}]$
Convolutional Blocks	Number of blocks	$\{2, 3\}$
	Filters (2 blocks)	NDP* from $\{32, 64, 128\}$
	Filters (3 blocks)	NDP* from $\{32, 64, 128\}$
	Activation	$\{\text{ReLU}, \text{ELU}\}$
	Pooling	$\{\text{Max}, \text{Avg}\}$
	Batch normalisation	$\{\text{True}, \text{False}\}$
	Dropout	$[0.0, 0.3]$
Dense Layers	Number of layers	$\{1, 2\}$
	Units (1 layer)	$\{128, 256\}$
	Units (2 layers)	NIP** from $\{128, 256\}$
	Activation	$\{\text{ReLU}, \text{ELU}\}$
	Dropout	$[0.2, 0.5]$

* NDP: Non-Decreasing Progression — filter counts are non-decreasing across blocks.

** NIP: Non-Increasing Progression — unit counts are non-increasing across layers.

Table 4: Optuna hyperparameter search — all trials

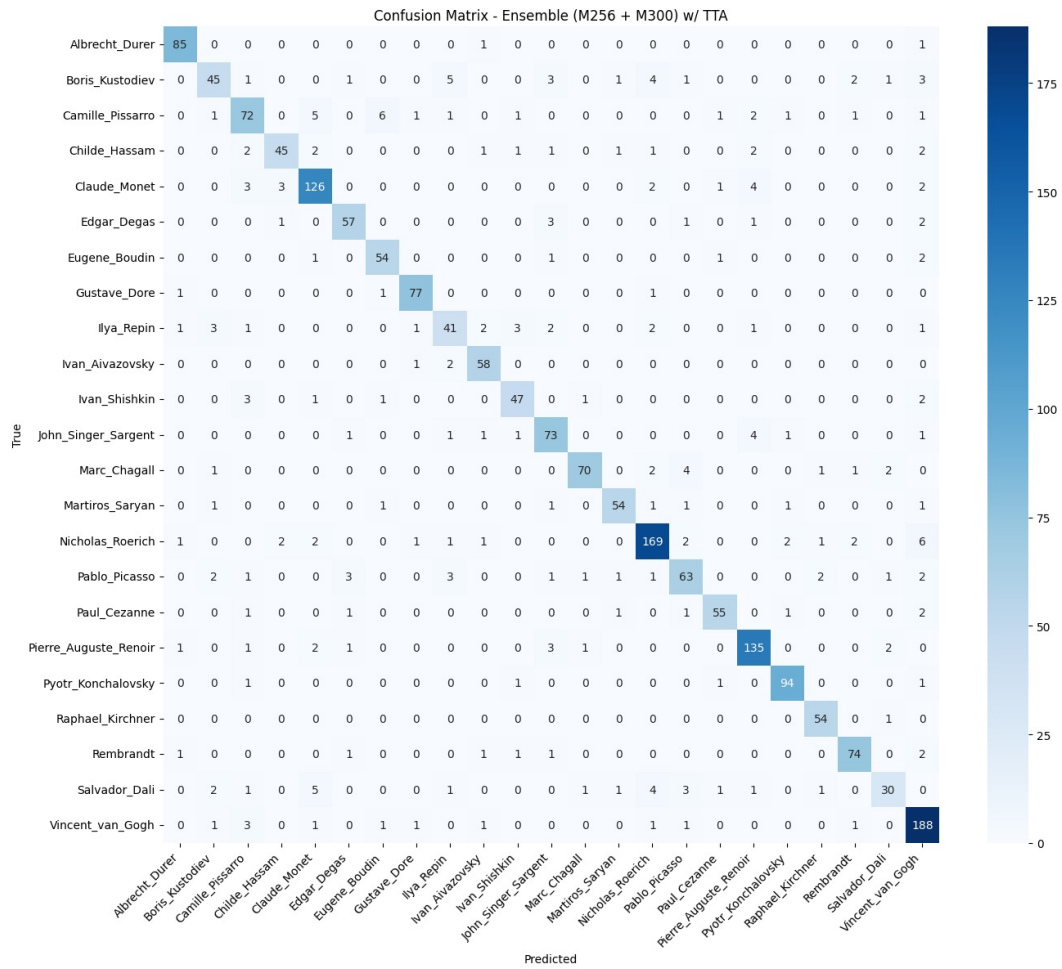
Trial	Filters	Conv Act.	Pooling	BN	Conv Drop.	Dense Units	Dense Drop.	LR	Epochs	Val F1
0	64→128	ReLU	Max	No	0.18	256→256	0.26	7.6e-4	35	0.3883
1	32→128	ReLU	Max	No	0.14	256→256	0.21	2.0e-3	35	0.3980
2	64→128	ReLU	Max	Yes	0.04	128	0.40	1.0e-3	20	0.3645
3	32→32→64	ReLU	Max	No	0.03	128	0.28	3.4e-3	35	0.4049
4	32→128	ELU	Avg	Yes	0.00	128→128	0.22	1.1e-3	35	0.4002
5	64→64→128	ReLU	Avg	Yes	0.27	128	0.37	2.9e-3	35	0.3382
6	32→64→128	ELU	Avg	No	0.27	128	0.22	9.7e-4	35	0.2549
7	64→128→128	ELU	Max	Yes	0.24	256→256	0.33	5.3e-3	35	0.3809
8	32→128	ReLU	Avg	Yes	0.16	256→128	0.28	4.0e-4	28	0.3676
9	32→32	ReLU	Avg	Yes	0.04	256	0.43	2.9e-3	22	0.3285
10	32→32→64	ELU	Max	No	0.09	256	0.50	4.9e-3	35	0.4037
11	32→32→64	ELU	Max	No	0.09	256	0.48	4.9e-3	35	0.3913
12	32→32→64	ELU	Max	No	0.09	256	0.49	5.0e-3	35	0.4120
13	32→32→64	ELU	Max	No	0.08	256	0.32	3.2e-3	35	0.4491
14	32→64→64	ELU	Max	No	0.09	256	0.33	5.4e-4	35	0.3702
15	32→64→64	ELU	Max	No	0.21	256	0.45	1.9e-3	35	0.3618
16	32→64→64	ELU	Max	No	0.12	256	0.36	2.2e-3	31	0.4061
17	32→64→128	ELU	Max	No	0.08	256	0.31	3.9e-3	35	0.4396
18	32→64→128	ELU	Max	No	0.06	256	0.31	2.4e-3	35	0.4318
19	64→64→128	ELU	Max	No	0.07	256	0.29	9.5e-3	35	0.4610
20	64→128→128	ELU	Max	No	0.13	128→128	0.25	9.2e-3	35	0.4222
21	64→64→128	ELU	Max	No	0.07	256	0.30	4.0e-3	35	0.4773
22	64→64→128	ELU	Max	No	0.06	256	0.30	3.7e-3	35	0.4685
23	64→64→128	ELU	Max	No	0.01	256	0.29	3.9e-3	35	0.4650
24	64→64→128	ELU	Max	No	0.01	256	0.25	4.1e-3	35	0.4639
25	64→64→128	ELU	Max	No	0.03	256	0.34	2.3e-3	35	0.4419
26	64→128→128	ELU	Avg	No	0.05	256	0.35	4.1e-3	8	0.0587
27	64→64→128	ELU	Max	No	0.01	256	0.38	2.7e-3	35	0.4476
28	64→64→128	ELU	Max	No	0.11	128	0.24	3.6e-3	35	0.4656
29	32→32	ELU	Max	No	0.17	256→128	0.24	6.1e-4	21	0.2296
30	64→128→128	ELU	Max	No	0.11	128	0.20	1.8e-3	34	0.4450
31	64→64→128	ELU	Max	No	0.05	128	0.28	3.5e-3	35	0.4643
32	64→64→128	ELU	Max	No	0.02	128	0.26	4.3e-3	35	0.4654
33	32→64→128	ELU	Max	No	0.11	128	0.26	4.4e-3	16	0.3177
34	64→64→128	ReLU	Max	No	0.14	128	0.23	2.8e-3	35	0.3409
35	32→64	ELU	Max	No	0.03	128	0.26	2.3e-3	35	0.3492
36	64→128→128	ReLU	Max	Yes	0.30	128	0.30	3.4e-4	30	0.3645
37	64→64→128	ELU	Avg	No	0.02	256→128	0.26	3.7e-3	23	0.1668
38	64→64	ReLU	Max	Yes	0.06	128	0.21	4.5e-3	27	0.4203
39	32→64→128	ELU	Avg	No	0.04	128	0.27	2.9e-3	35	0.3988

Appendix D: Test Time Augmentations

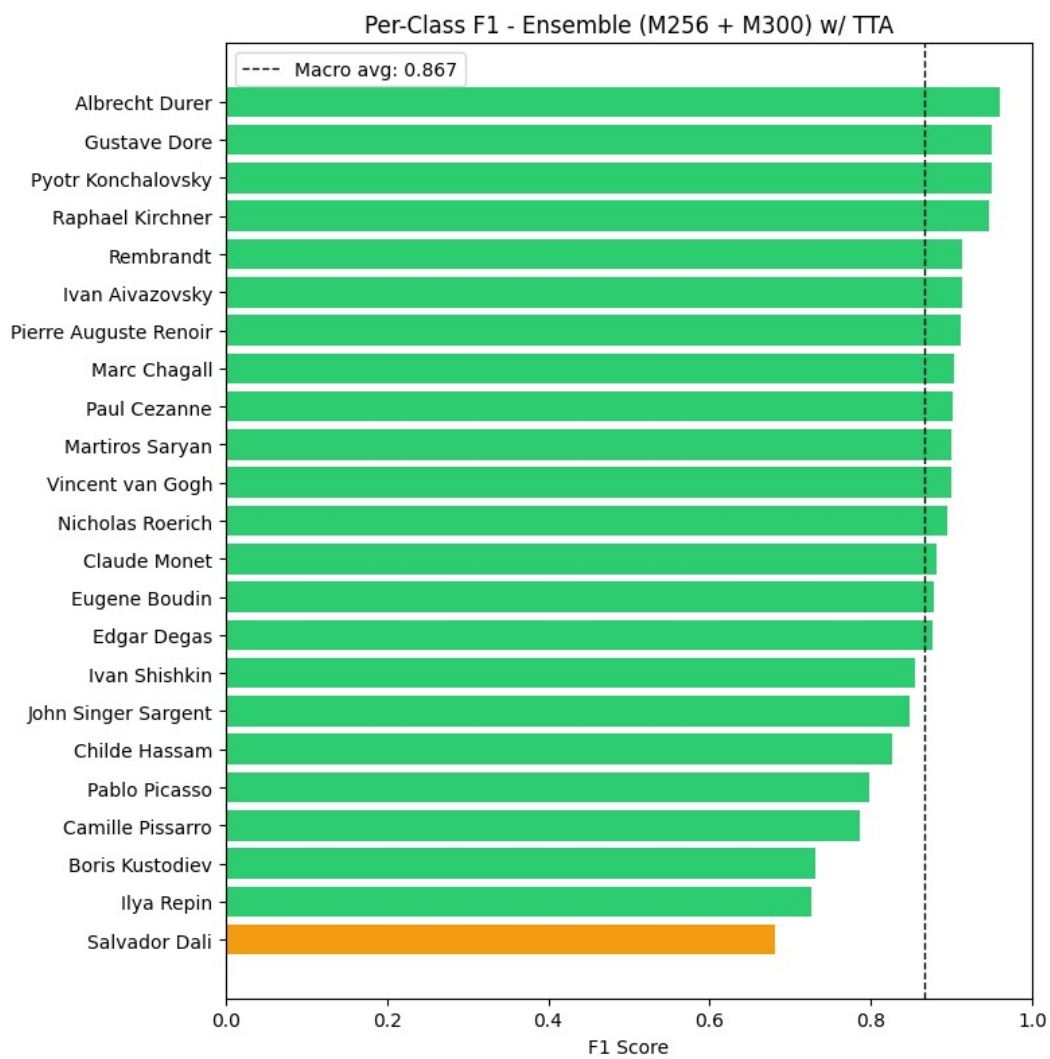
Table 5: Impact of TTA on accuracy and Macro F1-score per model.

Model	TTA Accuracy	TTA Macro F1	Δ Accuracy	Δ F1
Baseline Model	0.2753	0.1959	−0.0289	−0.0382
Optuna (Trial 21)	0.6004	0.5850	0.5258	0.5142
Residual Block	0.4483	0.3863	−0.2783	−0.3156
VGGNet	0.7371	0.7173	+0.0269	+0.0232
EfficientNetB3	0.7490	0.7334	+0.0227	+0.0245
EfficientNetB5	0.7758	0.7589	+0.0253	+0.0208
EfficientNetV2S	0.8743	0.8602	+0.0259	+0.0308
ResNet50	0.8519	0.8409	+0.0452	+0.0491

Ensemble Results with TTA



Model Evolution

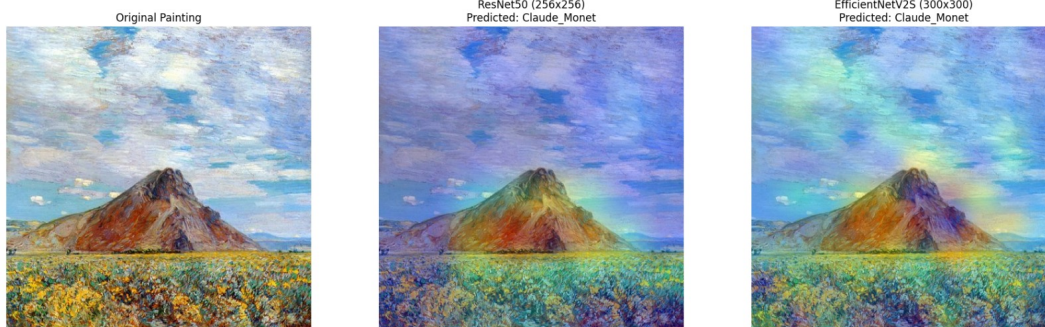


Training History

Appendix E: Grad-CAM Visualisations

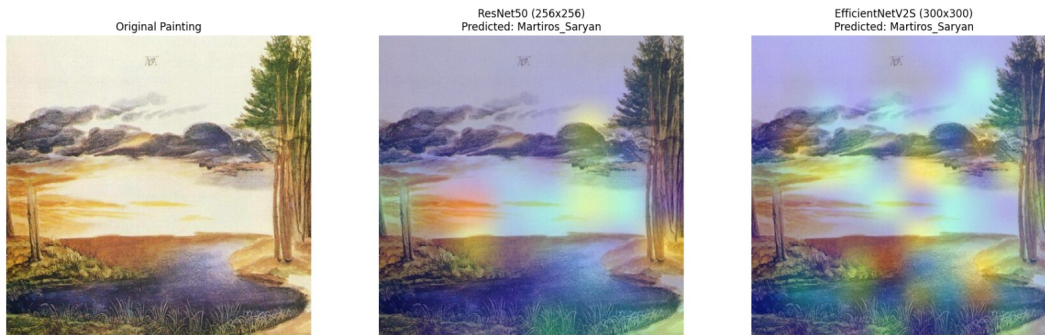
E.1 Example Visualisations

Ground Truth Label: True: Childe Hassam | Ensemble Pred: Claude Monet (91.9%)



Example 1

Ground Truth Label: Albrecht_Durer



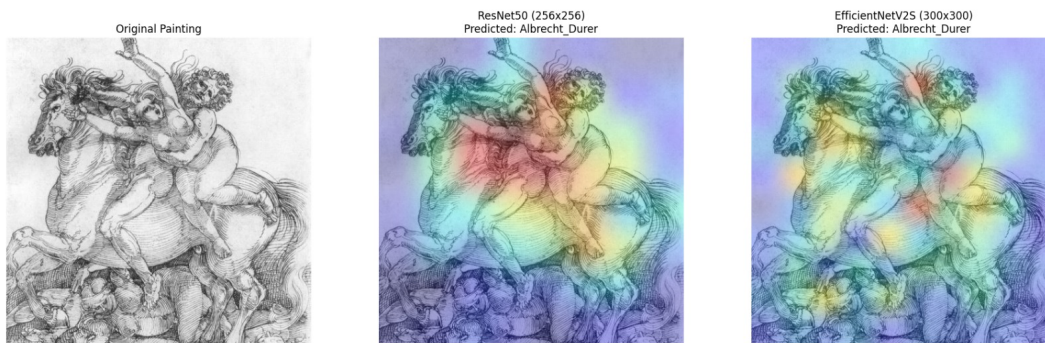
Example 2

Ground Truth Label: Albrecht_Durer



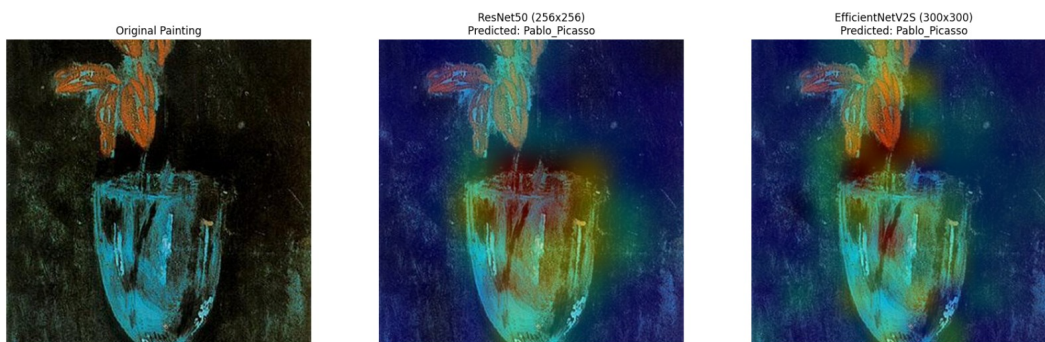
Example 3

Ground Truth Label: Albrecht_Durer



Example 4

Ground Truth Label: Pablo_Picasso



Example 5

E.2 Confident Correct and Incorrect Predictions

Most Confident Correct Classifications (Prototypical Styles)



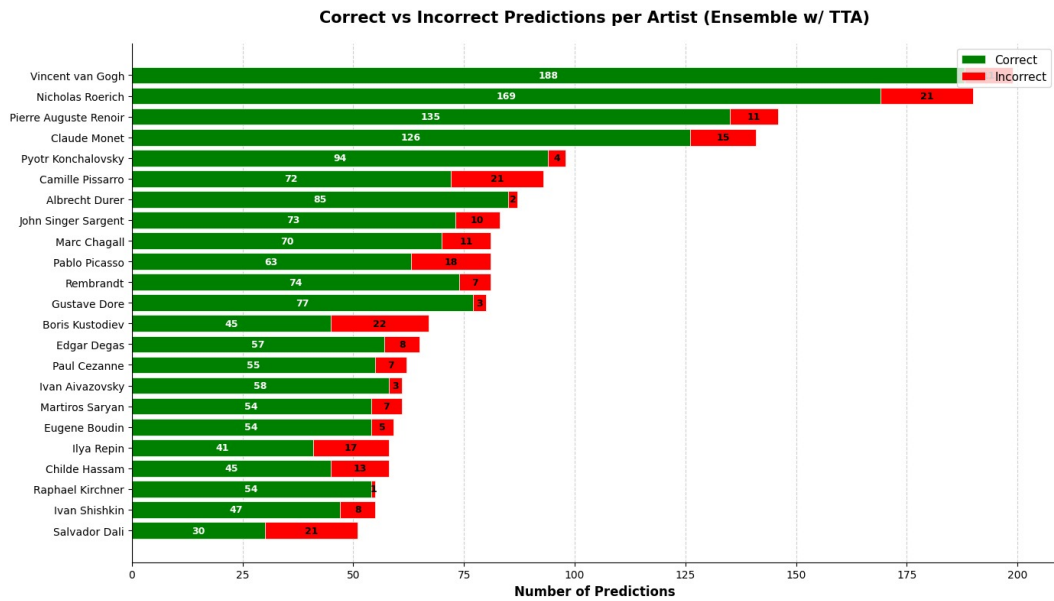
Confident Correct Predictions

Most Confident Misclassifications (Overconfident Errors)

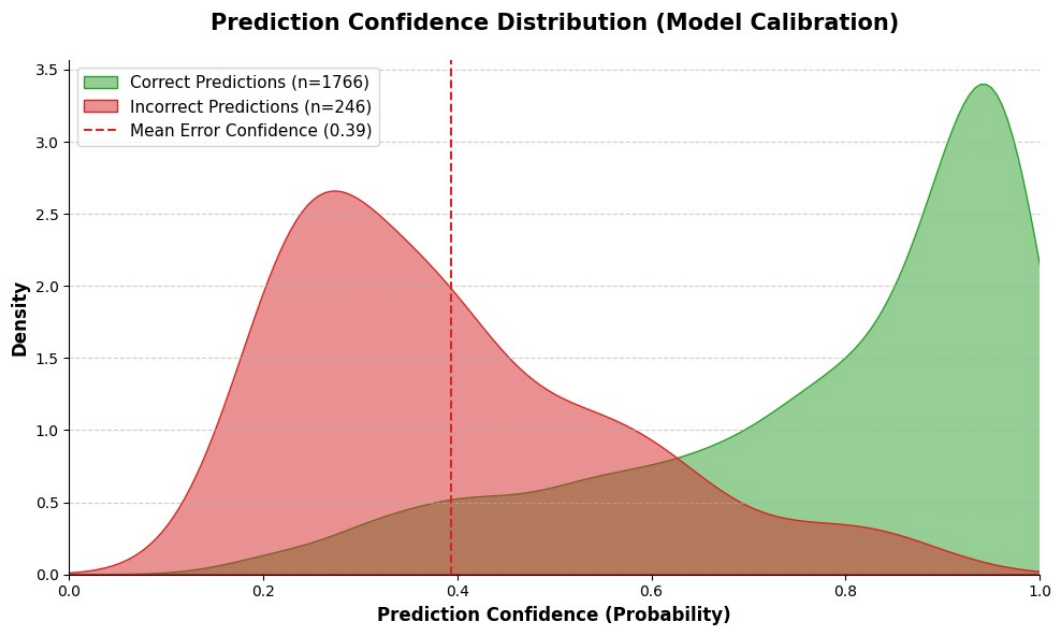


Confident Wrong Predictions

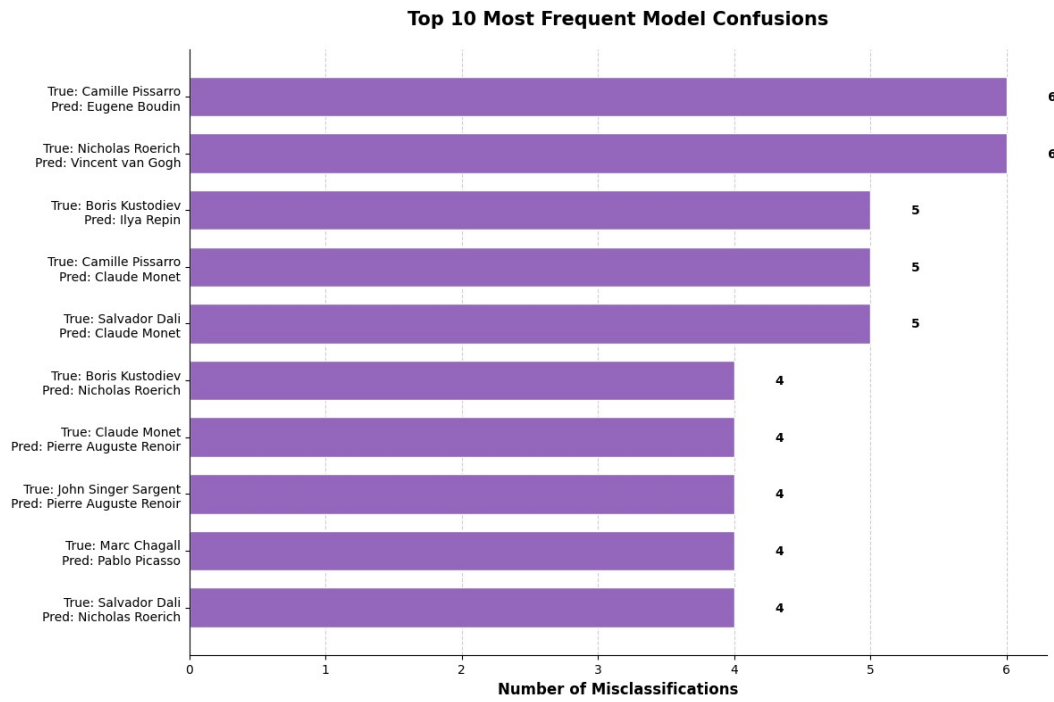
E.3 Additional Analysis



Correct vs. Incorrect Prediction Distribution



Model Calibration



Most Frequent Misclassifications